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Addressing Installation Challenges in Offshore Brownfield Projects: A Holistic Approach from Concept to Execution

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Abstract

This paper undertakes a critical examination of the construction and installation challenges inherently prevalent in offshore brownfield projects. Unlike greenfield developments, brownfield jobs involve integrating new infrastructure with existing, often aging, facilities, which introduces a layer of complexity that can significantly impede project timelines and escalate costs. The primary objective of this paper is to propose a comprehensive mitigation framework, meticulously designed to minimize offshore construction efforts and reduce overall project duration. Achieving this ambitious goal necessitates a paradigm shift towards a highly proactive and intricately integrated approach, commencing from the earliest stages of conceptual engineering, extending through meticulous construction at the fabrication yard, and culminating in the final offshore installation. Such an integrated methodology promises substantial benefits, not only for EPCI contractors by enhancing their efficiency and profitability, but also for Operators through diminished operational disruptions and reduced financial exposure. The aim is to transform the execution of brownfield projects into a more predictable, efficient, and cost-effective endeavor.

Introduction

The global energy sector, particularly in the Middle East, is undergoing a significant transformation. While the development of new hydrocarbon resources continues, there is an undeniable and growing emphasis on maximizing the value and extending the operational lifespan of existing offshore assets. This shift has led to a proliferation of brownfield projects—complex undertakings that involve the modification, upgrade, expansion, or refurbishment of operational offshore platforms. Unlike their greenfield counterparts, brownfield installations are inherently constrained by existing infrastructure, active production schedules, and the challenges of integrating new systems within a live, often aging, environment. This intricate interplay of operational continuity, structural limitations, and the dynamic offshore setting defines the unique challenges and opportunities of brownfield developments.

In the Middle East, the concentration of mature oil and gas fields necessitates a practical brownfield strategy to sustain production levels, enhance recovery, and adapt facilities to evolving operational demands

or new tie-back opportunities. Companies in the region are actively investing in these projects to unlock further reserves, improve operational efficiency, and ensure long-term asset integrity. NMDC Energy, a prominent player in the EPCI landscape, has notably been at the forefront of executing a significant number of such brownfield projects across the region. Their recent involvement underscores the escalating importance of these initiatives in the Middle East's energy strategy. However, the ambitious nature of these projects frequently encounters formidable hurdles, often compromising their timely and cost-effective execution.

Due to the fast-track nature of projects, the overall project lifecycle tends to be squeezed. Normally, installation aspects are ignored or not given the required attention in early stages of engineering. Another critical bottleneck for timely project execution, particularly in a region like the Middle East with significant offshore activity, is the unavailability of suitable vessels. Offshore brownfield projects often demand highly specialized marine assets, including jack-up construction barges & heavy-lift construction vessels, anchor handlers, diverse support vessels & material barges. These assets, especially large jack-up vessels with significant crane capacity are in high global demand, leading to tight scheduling, limited availability, and escalating charter rates. Any delay in project execution—whether due to engineering rework, late equipment delivery, or adverse weather—directly impacts vessel utilization, leading to higher costs and potentially forcing projects to wait for the next available slot, which could be months away. The selection of the appropriate vessel type and ensuring its timely availability with sufficient capacity are therefore paramount installation considerations that can impact a project's schedule and budget.

This technical paper aims to address the above challenges, drawing insights from recent NMDC Energy projects, and proposes a framework for their mitigation. By analyzing real-world scenarios and the strategies employed, or lessons learned, from these complex brownfield initiatives, we intend to illuminate best practices in project planning, design integration, and execution. The subsequent sections will investigate into specific mitigation strategies focusing on the critical areas of layout optimization, construction strategies, and vessel selection, demonstrating how a proactive, integrated approach can overcome these challenges, ensuring the safer, more efficient, and more predictable delivery of offshore brownfield installations. Through this analysis, we seek to provide valuable insights for future brownfield developments, reinforcing the importance of a holistic perspective to achieve project success in this demanding sector.

The Pitfalls of Fast-Track Project Execution

A significant prevailing challenge in modern offshore brownfield projects stems from their fast-track nature. Driven by market demands, economic pressures and the desire for rapid returns on investment, the overall project life cycle tends to be severely squeezed. This accelerated schedule often places immense pressure on engineering teams to expedite design phases, sometimes leading to crucial aspects being overlooked or given insufficient attention during the early, foundational stages of engineering.

Specifically, installation aspects are frequently ignored or not given the required attention in early stages of engineering. This oversight typically occurs because the primary focus during initial design is on functionality, structural integrity, and process integration. The intricate details of how large, complex modules will be transported, lifted, positioned and integrated offshore—including specific lifting points, transportation sea fastening, weight distribution during lift, required clearances, and landing interfaces—are often relegated to later stages or assumed to be the "installation contractor's problem." This deferred consideration of constructability and ability to install can lead to several severe and costly consequences such as:

- a. **Major Engineering Rework:** When installation challenges are identified late in the project cycle, perhaps during detailed design or even during fabrication, it could potentially necessitate significant engineering rework. Designs that were deemed complete may require extensive modifications to accommodate practical lifting limits, vessel capabilities, access constraints, or interface issues

with the existing platform. This rework is not merely a drafting exercise; it often involves re-performing structural analyses, re-drawing complex P&IDs or electrical schematics, and re-validating equipment selections. Such late-stage changes are inherently more expensive and time-consuming than addressing them proactively during initial design.

- b. **Wastage of Material and Resources:** Engineering rework directly translates into the wastage of both materials and resources. Components fabricated to an initial design might be rendered unusable or require costly modifications. Man-hours spent on initial, unfeasible designs are effectively lost. This extends beyond tangible materials to include wasted engineering effort, planning time, and even potential disruption to fabrication schedules as new materials are ordered or existing ones are re-processed. Such inefficiencies contribute directly to escalating project costs and environmental footprints.
- c. **Delays:** Perhaps the most immediate and impactful consequence of neglected installation planning is project delay. Late-stage engineering rework, the procurement of new or modified materials, and the need to devise ad-hoc solutions for unforeseen installation challenges inevitably push back critical milestones. These delays can ripple throughout the entire project schedule, impacting subsequent construction, hook-up, commissioning, and ultimately, the vital RFSU date. In projects where daily deferral costs are substantial, even minor delays can lead to financial penalties, far outweighing any perceived initial savings from rushing the engineering phase.

Therefore, to mitigate these critical risks, a proactive approach that deeply integrates installation considerations into the earliest phases of project engineering is indispensable. This sets the stage for optimized methodologies that address the complexities of brownfield environments head-on.

Proactive Planning & Early Engagement Strategies

To effectively counter the pressures of fast-track projects and avoid the costly pitfalls of late-stage installation challenges, robust planning and early engagement strategies are necessary. This involves a shift from reactive problem-solving to proactive foresight, ensuring that installation constraints and requirements are embedded in the project's foundational stages.

After the preliminary definition of the project layout and scope, several key elements must be immediately and thoroughly addressed:

Finalization and Advance ordering of all Long-Lead Equipment

Identifying and finalizing specifications for all long-lead equipment is a critical early step. This includes items with extensive manufacturing times, such as large compressors, turbines, specialized vessels, launchers/receivers or specific electrical components. Once identified, these items must be ordered well in advance to prevent them from becoming bottlenecks that dictate the entire project schedule. Delays in long-lead item procurement can ripple through engineering, fabrication, and ultimately, installation, making early commitment essential. This is a procurement strategy yields immediate engineering benefits by providing early data such as weights and dimensions, which is essential for developing transport and installation methodologies of those equipment.

Early Identification and Finalization of Lift Vessels

Offshore brownfield projects frequently involve significant heavy lifting operations. These may include installing new modules or equipment, replacing existing components, or lifting decommissioned parts. Lift vessels—either a Jack up lifting vessel or floating crane vessels in some cases, are highly specialized assets with extremely limited global availability. Therefore, early engagement and the finalization of the primary lift vessel are non-negotiable. This early commitment allows the project to secure the necessary asset, understand its specific capabilities and limitations, and integrate these directly into the engineering design.

Establishment of Baseline Lift Parameters

Once the target lift vessel is identified, critical baseline parameters for all major lifts must be established and adhered to throughout the detailed engineering phase. The lift parameters such as minimum required lifting capacity, lift radius and minimum hook heights for the lift become non-negotiable design constraints. These parameters dictate the module split, weight distribution, structural design, and even the platform layout. Fixing these baselines early prevents the design from progressing in a direction that might exceed the selected vessel's capabilities or require an infeasible lift configuration, thereby mitigating the risk of major engineering rework later. By committing to these parameters from the outset, engineering teams are provided with clear boundaries within which to design, ensuring constructability and align ability with the chosen installation strategy. It also allows optimizing the lifts and combining the lifts should thus fall within the vessel capability.

Early Identification of the transport vessel

Even though cost effective; conventional non propelled barges not usable either due to safety constrain near complexes or the inability to moor alongside Jack-Up vessels. This will lead search for offshore support vessels with large deck area and capacity to carry the intended modules. Given the limited availability of such large deck area/capacity vessels, early identification is essential.

By implementing these early engagement and planning strategies, project teams can proactively address potential installation challenges, select the most appropriate marine assets and streamline the entire project lifecycle. This integrated approach fundamentally improves the chances of on-time, on-budget and safe project delivery.

Engineering Optimizations

The Concept/FEED (Front-End Engineering Design) stage for offshore brownfield installations is an important phase of project success while generally EPCI contractors have no control over this phase. It's during this phase that critical decisions are made that will dictate the entire project lifecycle, from detailed design and procurement to fabrication, installation and commissioning. The inherent complexities of working on existing, live infrastructure in a marine environment demand a thorough and diligent approach. A key driver for these projects is often the desire to minimize offshore man-hours, which are inherently more expensive, logistically challenging and carry higher safety risks compared to onshore activities. The following paragraphs suggest some practical methodologies for optimizing the layout of new facilities: deck extensions and multi-deck modules, exploring their benefits, intricate challenges and the detailed considerations required during the Concept/FEED stage to ensure their successful implementation.

New Facilities Arranged with Deck Extensions: Maximizing Onshore Pre-Fabrication

The strategy of incorporating deck extensions involves designing new areas that protrude from or are integrated into the existing platform's perimeter. These extensions are typically fabricated onshore as complete or near-complete units, including structural elements, piping, electrical systems and pre-installed equipment.

Arranging Deck Extensions in Multiple Deck Levels

A highly effective strategy for offshore projects involves pre-assembling and installing multi-level deck extensions as single, integrated units. This method fundamentally changes the traditional "stick-built" approach, which involves installing individual decks and components piece by piece. Instead, the entire multi-deck structure is fabricated in a controlled onshore environment before being transported and lifted as a single, complete module. By shifting to this integrated strategy, projects can significantly reduce offshore exposure and leverage the advantages of onshore fabrication.

This pre-assembly approach greatly improves a project's safety, quality, and efficiency. It moves the majority of the fabrication work from the high-risk offshore environment—where personnel are exposed to harsh weather and working at height—to a safer onshore facility. This controlled setting allows for superior quality control, enabling precise welding, painting, and insulation. Any design flaws or fabrication errors can be identified and corrected onshore much more cost-effectively than they could be offshore, which minimizes the need for expensive rework. This shift also delivers substantial gains in cost and schedule by using lower labor rates and dedicated machinery.

Below [figure 1](#) refers an ongoing project multilevel extension which was initially planned as three individual deck installation. Design was optimized to 3 level single lift installation module. By shifting to a multi-level lift strategy, the project significantly reduces offshore exposure and leverages the advantages of onshore fabrication.

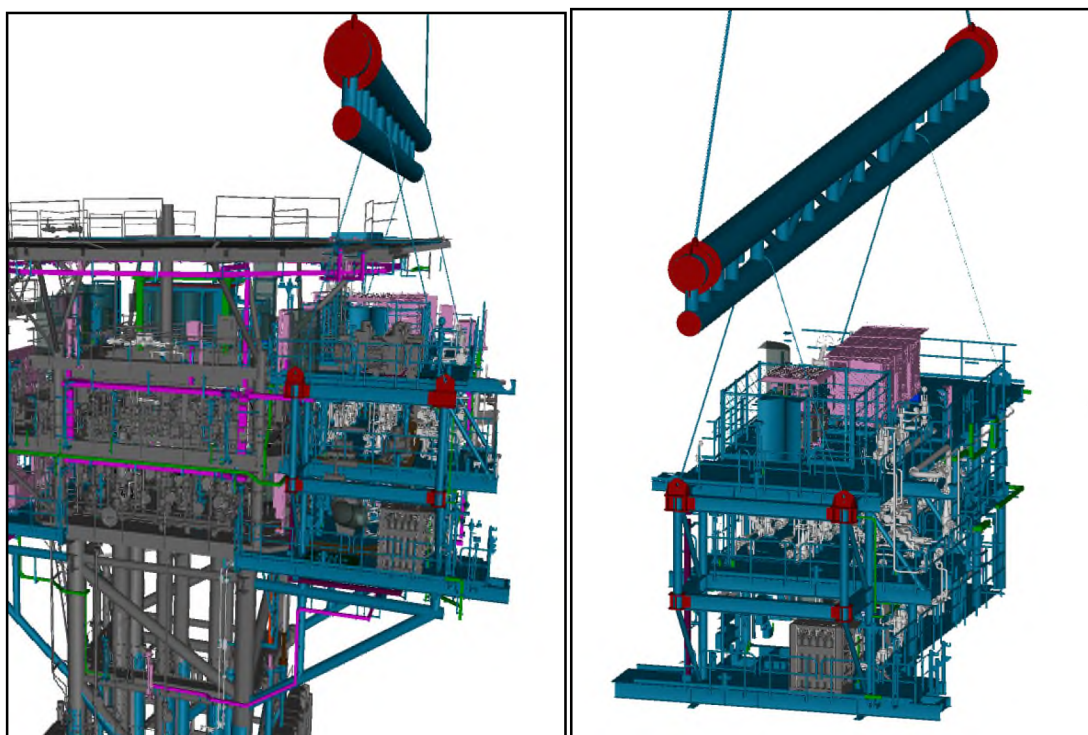


Figure 1—A multilevel Deck extensions Arrangement & Installation

Extensions Underneath Platforms - Optimizations

Designing under-deck extensions on offshore platforms presents unique challenges, particularly when aiming for efficient and safe installation. [Figure 2](#) shows a recent project where an initial design for an under-deck extension involved a slender panel. While structurally not sound, this configuration made installation as a single module practically impossible due to its geometry and the limited access points under the platform. Recognizing this critical installation bottleneck, the design team initiated a significant re-evaluation.

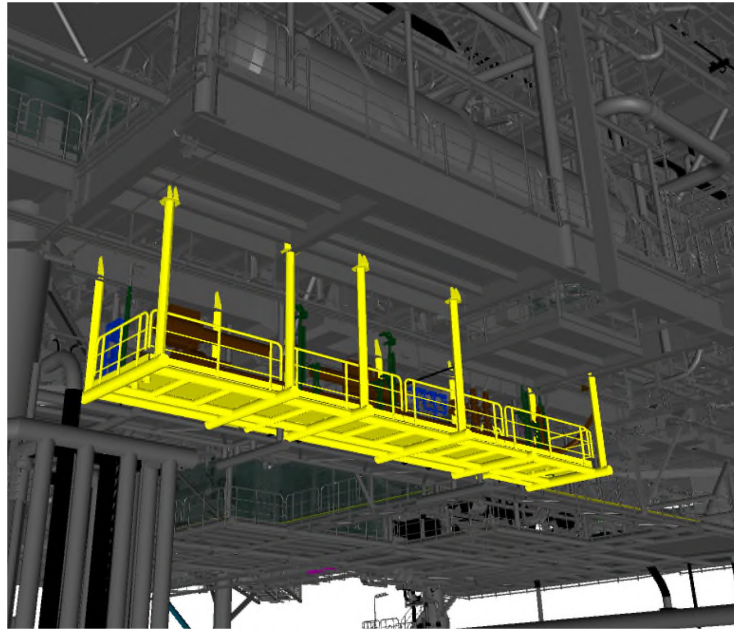


Figure 2—Extension under platform before installation optimizations

The solution involved a strategic redesign where the extension was conceptualized as a wider & shorter panel with a strategically shifted location closer to edge of the platform (Figure 3). This optimized design wasn't just about changing dimensions; it was about integrating the installation methodology directly into the fabrication plan. With this re-engineered approach, the module is now specifically designed to be brought into its precise position using dedicated trunnions. These integrated lifting points allow for controlled movement and alignment, making it feasible to install the entire extension as a single, prefabricated module. This thoughtful optimization directly addresses the core aim: vastly improving the ease of installation and significantly reducing the "crawling distance" and manual effort required offshore. Figure 4 indicates proposed installation scheme with minimum crawling distance.

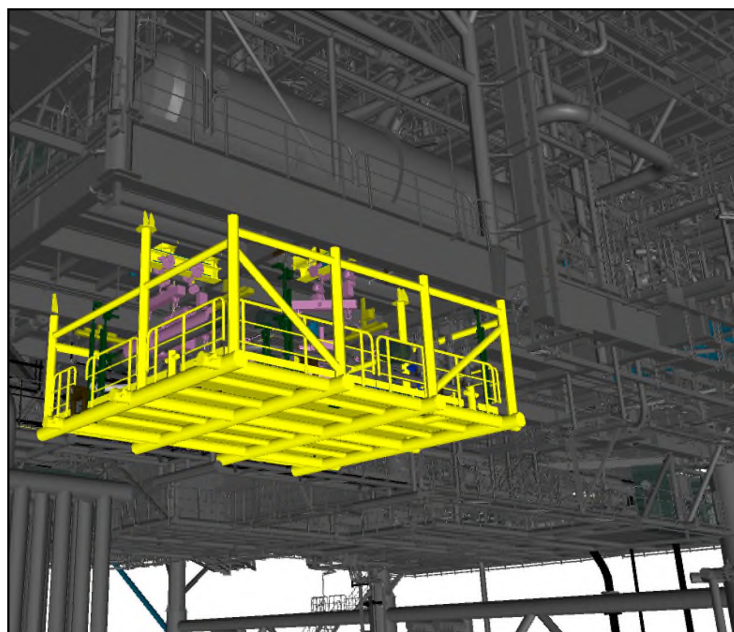


Figure 3—Extension under platform—design and location optimized for installation

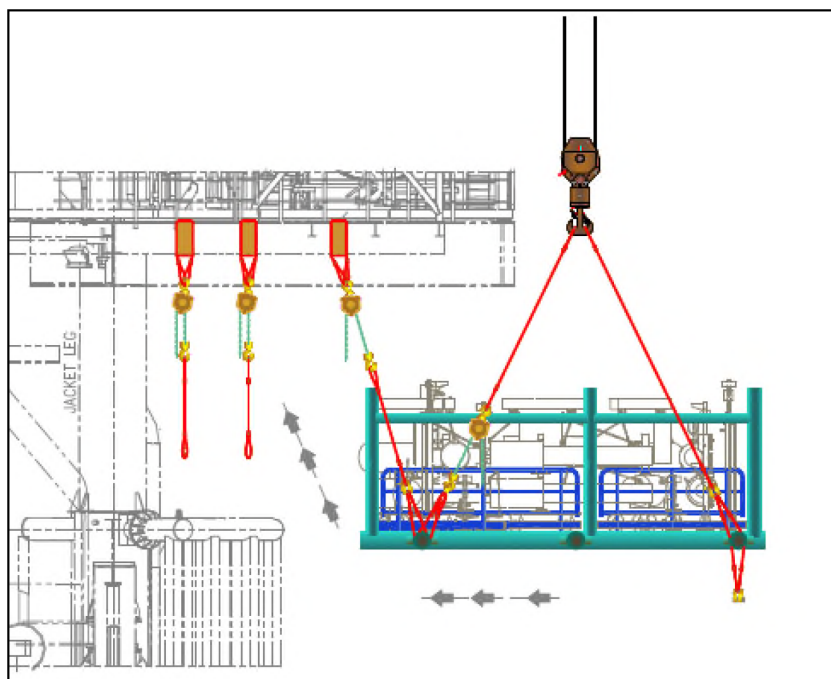


Figure 4—Extension under platform-Installation scheme

Offshore Activities

Below paragraphs explains some of the methodologies adopted by NMDC energy for optimizations on offshore activities:

The Phased Installation Approach: Pre-Installed Support Frames

The principle of this optimized strategy centers on the strategic consideration and utilization of pre-installed supports for larger modules. This methodology dictates that the structural reception frames, designed to receive the new modules, are fabricated and installed.

This crucial preparatory phase encompasses a range of activities, including:

- a. **Foundation Preparation:** Ensuring the existing platform structure is adequately prepared to receive the new support frames. This may involve minor structural strengthening, cleaning, or precise alignment checks to guarantee a stable foundation.
- b. **Installation of Reception Frames:** Critical task of lifting and securing the prefabricated support frames onto their exact designated positions on the platform. This often involves complex rigging, extremely precise positioning, and welding to the existing structure. The installation of these support frames also has its own challenges, particularly given their intended placement directly onto jacket legs. These structural elements are frequently characterized by significant congestion from existing risers and piping infrastructure, which results in extremely restricted installation clearances. Below [figure 5](#) shows a typical example in recently executed project with restricted space constrains for installation

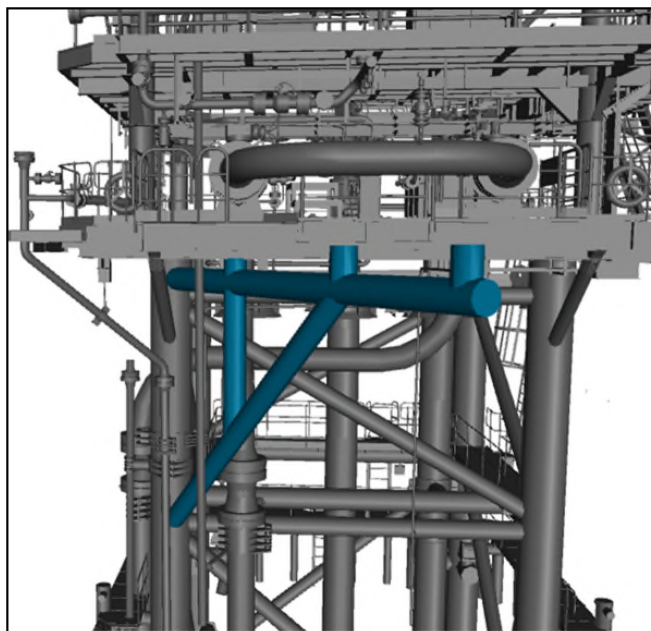


Figure 5—Module support frame with restricted space constraints

To precisely navigate these spatial constraints and ensure accurate placement, specific techniques are employed. Pre-installed guide pins are meticulously integrated into the jacket structure. These pins serve as critical alignment features, accurately guiding the support frame during its descent and ensuring its exact engagement with the designated welding points on the jacket. Below [figure 6](#) shows installation methodology adopted for above structure.

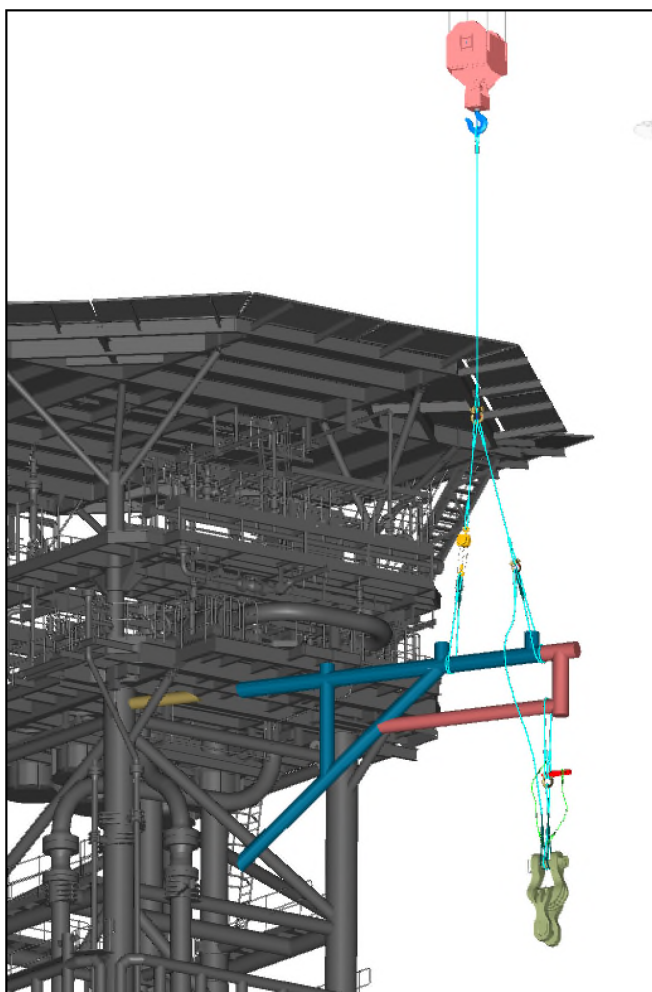


Figure 6—Module support frames installation with help of guide pin

Furthermore, counterweights are often deployed on the opposing side of the lift to manipulate the module's Center of Gravity (CoG), ensuring it remains outside the immediate confines of the platform's existing structure. This engineered solution facilitates a controlled, direct engagement of the support frame with its installation interface, preventing undesirable dynamic oscillations during final placement. This methodology ensures a stable and predictable landing, even amidst the tight clearances and complex existing infrastructure of a brownfield site. The [figure 7](#) shows a typical platform with support frames installed for multiple extensions.

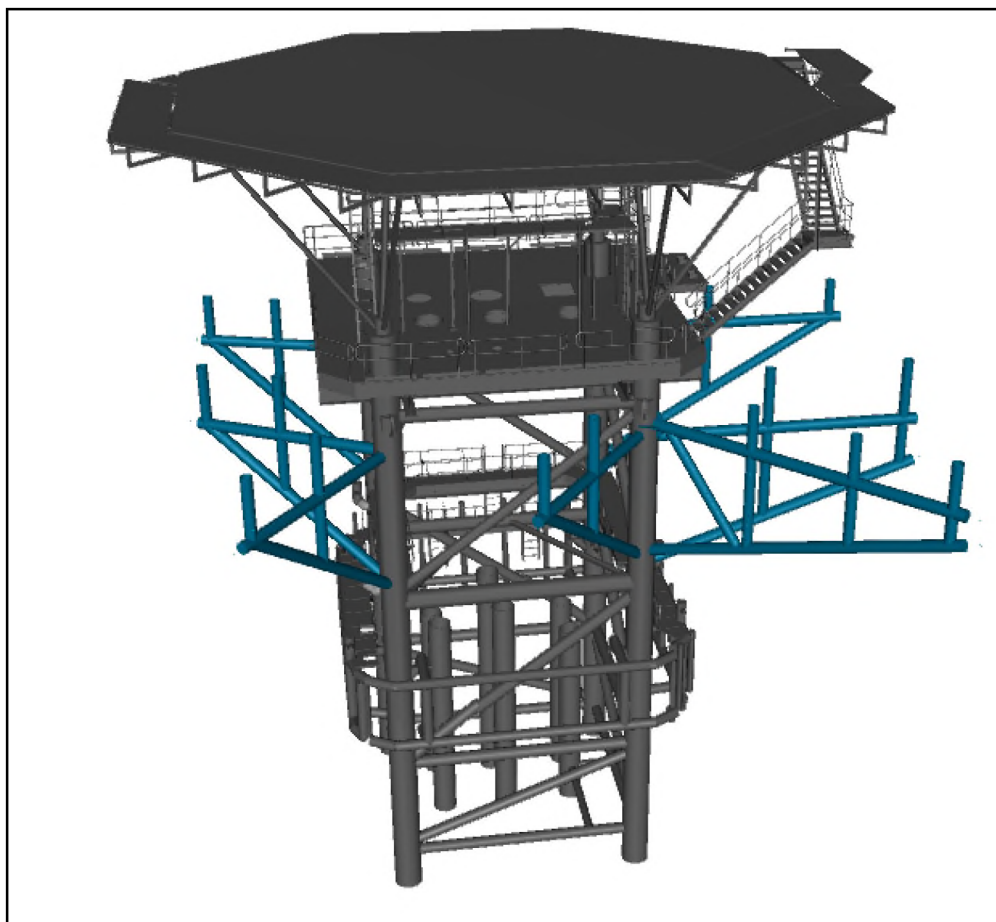


Figure 7—platform with pre-installed reception frames for multiple extension

Use of Multiple Vessels

For offshore platforms requiring the installation of large, heavy modules over an extended duration, a strategic approach involving different vessel types offers significant advantages in efficiency, safety, and cost-effectiveness. This multi-vessel strategy optimizes each phase of the installation process.

Initially, smaller jack-up barges are ideally suited for preparatory works. A key enabler for this initial preinstallation phase is the deployment of a smaller capacity jack-up barge for preparatory works. These smaller barges are a strategic choice because they have significantly lower day rates compared to the large Jack up vessel or heavy-lift vessels needed for the primary module installation. Furthermore, they offer greater flexibility in terms of operational windows, allowing for more extended periods of preparatory work.

Subsequently, for the installation of the larger, heavier modules, more specialized vessels become indispensable. Dedicated jack-up vessels with superior lifting capacity, or floating crane vessels, are deployed for this phase. The intention is to have cost effective usage of vessels based on requirements.

Strategic Selection of Installation Vessel: Jack-Up Lifting Barge vs. Floating Heavy-Lift Vessels

For the installation of modules weighing approximately 100 metric tons or more, the selection of an appropriate installation vessel, specifically between a traditional floating heavy-lift vessel and a jack-up lifting barge, is a critical strategic decision.

Traditionally, EPC contractors have often favored heavy-lift floating vessels for larger modules due to their bigger lifting capacity and suitability for open-water operations. However, the deployment of floating vessels, particularly in brownfield environments, presents several challenges such as:

- a. **Motion Sensitivity:** Floating vessels are susceptible to wave motion and swell, which can significantly restrict lifting windows and compromise precision. Even moderate sea states can render lifting operations unsafe, leading to weather delays and increased operational risk.
- b. **Limited Stability for Precision and Tight Clearances:** Brownfield platforms typically feature constrained installation clearances. Guidelines requires a 3-meter horizontal clearance for the lifted object from existing structures if not specially designed guides and bumpers are installed^(Ref 1). The implementation of such guides and bumpers is often challenging in brownfield platforms due to inherent space limitations and complexities. Consequently, the potential use of a floating lift vessel necessitates early identification during the engineering phase to incorporate required layout modifications.
- c. **Higher Operational Costs in Adverse Conditions:** Weather-induced delays directly increase the cumulative vessel days, leading to significant unforeseen project costs.

In contrast, jack-up lifting barges offer a compelling alternative for module installation in many brownfield scenarios, providing distinct advantages:

- a. **Stability and Precision:** Once jacked up, these barges create a stable, land-like platform, isolated from wave motion. This provides stability and precision for lifting and positioning heavy modules, even in less favorable weather conditions. This significantly reduces fit-up challenges and the need for complex offshore rework.
- b. **Reduced Installation Clearances and Collision Risk:** Jack-up barges generally require comparatively less installation clearance. Their static nature, once jacked up, also reduces the risk of accidental contact with existing platform infrastructure during complex lifts compared to a dynamically positioned/moored floating vessel.
- c. **Extended Weather Windows:** The inherent stability of jack-up barges allows for a broader operational window, enabling lifts to proceed in sea states that would typically halt a floating vessel. This predictability can substantially compress the offshore installation schedule.

While jack-up barges may have limitations in very deep waters or situations requiring extreme transit speeds, their stability and precision generally outweigh these factors for the typical water depths and operational requirements of brownfield projects, especially in regions such as the Middle East. Their capacity to deliver modules accurately and efficiently, minimizing weather downtime and offshore rework, positions them as a strategically advantageous choice for managing the inherent complexities and costs associated with integrating new infrastructure into existing platforms. The strategic selection of the appropriate installation vessel is paramount to ensuring safer, more efficient, and more predictable project delivery.

Transport Options: Conventional Barges and Offshore Support Vessels

Traditionally, barges have been the standard for transporting modules. However, their use presents significant limitations in congested offshore fields. A primary challenge is that conventional anchoring is often prohibited due to a lack of space. When using jack-up barges, the elevation difference makes it impossible to use conventional barge-to-barge mooring.

In some of the complexes, the use of non-propelled supply barges is restricted due to safety concerns.

To overcome these significant constraints, the industry has developed and refined alternative strategies. One common solution is to employ offshore support vessels that can use specific, permitted anchor patterns designed for tight spaces. In the most confined scenarios—where even a single forward anchor cannot be deployed—the only viable solution is to use a dynamically positioned (DP) vessel.

Choosing a supply vessel over a conventional cargo barge offers several key advantages:

- a. Improved Operability and Faster Turnaround: Supply vessels are more agile and can complete tasks more quickly.
- b. Better Maneuverability: Their design allows for easier navigation in tight spaces. Dynamic Positioning System: DP capabilities enable precise station-keeping without the need for anchors.
- c. Dynamic Positioning System: DP capabilities enable station-keeping without the need for anchors in congested fields.

Despite these benefits, this option also comes with some notable drawbacks:

- a. Lesser Capacity: It can be difficult to find a suitable vessel for modules weighing approximately 100Mt or more
- b. Increased Trips: Lower capacity means multiple trips are often required, which can be less efficient than a single barge trip.
- c. Severe Motion Responses: Supply vessels are more susceptible to wave motion than barges, which can affect stability during operations.

To carry large modules/equipment support vessel with larger deck area and deck strength will be required. Availability of such vessels will be limited, so early engagement will be necessary. While cargo barges are often considered as the cheaper option on a day-rate basis, a comprehensive project-specific analysis is essential. This evaluation must consider a wide range of factors, including the module's weight and dimensions, the number of trips required, the environmental conditions of the specific offshore field, and most importantly, the regulatory and safety requirements for each intended lift. By assessing these variables, EPCI contractors can select the most suitable vessel, ensuring a safe, efficient, and cost-effective installation.

Conclusion: Forging a Predictable Path for Offshore Brownfield Project Success

This paper provides a critical examination of the challenges in offshore brownfield projects, arguing that a shift from reactive to proactive project management is essential. The traditional "fast-track" approach often delays installation planning, leading to costly engineering rework, material wastage, and schedule delays. To address these pitfalls, a holistic framework is proposed that embeds installation considerations from a project's earliest stages. This proactive planning, including the early finalization of long-lead equipment and lift vessel selection, provides clear design constraints to prevent unfeasible solutions and ensure a more predictable project delivery. Furthermore, engineering optimization during the Concept/FEED phase is very critical this approach. Strategies like designing multi-level deck extensions for single-lift installation and re-engineering under-platform modules for easier access and reduce offshore man-hours. This strategic shift leverages the controlled environment of onshore fabrication yards, improving quality, enhancing safety, and accelerating the project schedule by minimizing "stick-build" work offshore.

The paper also outlines an approach to offshore activities by advocating for the use of multiple vessels for platforms with large extend of modifications. Moreover, the strategic selection of a jack-up lifting barge over a traditional floating vessel for module installation and strategic selection of transport vessel meeting essential operational and safety requirements.

In conclusion, the successful execution of offshore brownfield projects, particularly in a demanding region like the Middle East, hinges on a proactive and deeply integrated project philosophy. By meticulously addressing installation considerations from the Concept/FEED stage, leveraging onshore pre-fabrication, employing innovative phased installation methodologies, and making strategic vessel choices, the industry can transform these inherently complex undertakings into predictable, efficient, and cost-effective endeavors, thereby ensuring long-term asset integrity and accelerated returns on investment. This

holistic approach, as highlighted by recent industry experiences, is a practical optimization to reduce the project life cycle of brownfield projects.

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